



**CSE 422 Lab Project: Predicting Academic Outcomes Using Machine Learning: A
Comparative Study of Classification Algorithms**

**Sammam Mahdi
Student ID: 22201628**

**Reshad Ul Karim
Student ID: 22201594**

BRAC University

Course: CSE422: Artificial Intelligence
Instructors: Farhan Faruk, Aunon Halder
Department of Computer Science and Engineering
BRAC University

Date: May 8, 2025

Table Of Contents

1. Introduction	3
2. Dataset	3
2.1 Dataset description	3
2.2. Exploratory Data Analysis (EDA)	4
2.2.1 Descriptive Stats	4
2.2.2 Descriptive Stats: Summary Statistics of Numerical Features	5
2.2.3 Skewness Interpretation	6
2.2.4 Summary Statistics of Categorical Features	7
2.5 Class Distribution	8
2.6 Distribution and Outlier Analysis	10
2.7 Correlation Matrix Analysis	12
3. Dataset Pre-processing	14
3.1 Fault: Missing Target Labels	14
3.2. Fault: Irrelevant or Redundant Features	14
3.3 Fault: Mixed Data Types & Categorical Codes	15
3.4. Fault: Missing Values in Predictors	16
3.5. Fault: Train/Test Leakage & Split	17
Neural Network	18
Random Forest	19
XGBoost	20
K-Nearest Neighbors	21
Decision Tree	22
Logistic Regression	23
Discussion	24
Conclusion	26

1. Introduction

Predicting academic outcomes such as graduation, dropout, or continued enrollment is a critical task in educational data mining. Accurate predictions can help universities identify at-risk students early and take proactive measures to improve retention and support services. This project focuses on building and evaluating machine learning models to classify students into three categories—Graduate, Dropout, and Enrolled—based on demographic, academic, and socio-economic features. We apply several classification algorithms, including Neural Networks, Random Forest, Extreme Gradient Boosting (XGBoost), K-nearest neighbour (KNN), Decision Tree, and Logistic Regression, to determine which approach performs best. The study emphasizes data preprocessing, class imbalance handling, and thorough evaluation using metrics such as accuracy, precision, recall, and F1-score.

2. Dataset

2.1 Dataset description

- i. The dataset contains **24 features**.
- ii. Based on the ‘Target’ values given, we can see that it’s a classification problem as the samples in the dataset can be classified into 3 categories – (‘Dropout’, ‘Graduate’, and ‘Enrolled’).
- iii. The dataset contains a total of 4424 samples or datapoints.
- iv. The dataset contains a mixture of quantitative and categorical features.

Sl	Feature Type	Feature Name	Sl	Feature Type	Feature Name
1	Categorical	Marital status	13	Categorical	Debtor
2	Categorical	Application mode	14	Categorical	Tuition fees up to date
3	Categorical	Course	15	Categorical	Gender
4	Categorical	Daytime/evening attendance	16	Categorical	Scholarship holder
5	Categorical	Previous qualification	17	Categorical	International
6	Categorical	Nacionality	18	Numerical	Application order

7	Categorical	Mother's qualification	19	Numerical	Previous qualification (grade)
8	Categorical	Father's qualification	20	Numerical	Admission grade
9	Categorical	Mother's occupation	21	Numerical	Age at enrollment
10	Categorical	Father's occupation	22	Numerical	Unemployment rate
11	Categorical	Displaced	23	Numerical	Inflation rate
12	Categorical	Educational special needs	24	Numerical	GDP

2.2. Exploratory Data Analysis (EDA)

2.2.1 Descriptive Stats

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
Marital status	4001	NaN	NaN	NaN	1.172457	0.5846	1	1	1	1	6
Application mode	3941	NaN	NaN	NaN	18.56204	17.4532	1	1	17	39	57
Application order	3998	NaN	NaN	NaN	1.730865	1.31068	0	1	1	2	9
Course	3959	NaN	NaN	NaN	8856.62895	2063.15	33	9085	9238	9556	9991
Daytime/evening attendance	3984	NaN	NaN	NaN	0.892319	0.31002	0	1	1	1	1
Previous qualification	3990	NaN	NaN	NaN	4.525564	10.1327	1	1	1	1	43
Previous qualification (grade)	3952	NaN	NaN	NaN	132.690056	13.2266	95	125	133	140	190
Nacionality	3978	NaN	NaN	NaN	1.860985	7.04042	1	1	1	1	109

Mother's qualification	4010	NaN	NaN	NaN	19.482294	15.6205	1	2	19	37	44
Father's qualification	3974	NaN	NaN	NaN	22.20307	15.3821	1	3	19	37	44
Mother's occupation	3988	NaN	NaN	NaN	10.895687	26.3048	0	4	5	9	194
Father's occupation	3999	NaN	NaN	NaN	10.910978	25.0033	0	4	7	9	194
Admission grade	3962	NaN	NaN	NaN	126.903609	14.5069	95	118	126	135	190
Displaced	3982	NaN	NaN	NaN	0.550477	0.49751	0	0	1	1	1
Educational special needs	3976	NaN	NaN	NaN	0.011569	0.10695	0	0	0	0	1
Debtor	3972	NaN	NaN	NaN	0.114804	0.31883	0	0	0	0	1
Tuition fees up to date	3998	NaN	NaN	NaN	0.882941	0.32153	0	1	1	1	1
Gender	3987	NaN	NaN	NaN	0.352646	0.47785	0	0	0	1	1
Scholarship holder	3974	NaN	NaN	NaN	0.251132	0.43372	0	0	0	1	1
Age at enrollment	3980	NaN	NaN	NaN	23.227638	7.60824	17	19	20	25	70
International	3996	NaN	NaN	NaN	0.025526	0.15774	0	0	0	0	1
Unemployment rate	3995	NaN	NaN	NaN	11.575394	2.66562	7.6	9.4	11.1	13.9	16.2
Inflation rate	4002	NaN	NaN	NaN	1.228286	1.38086	-0.8	0.3	1.4	2.6	3.7
GDP	3968	NaN	NaN	NaN	0.017656	2.27438	-4.1	-1.7	0.32	1.79	3.51
Target	3971	3	Graduate	1979	NaN	NaN	NaN	NaN	NaN	NaN	NaN

2.2.2 Descriptive Stats: Summary Statistics of Numerical Features

	count	mean	std	min	25%	50%	75%	max
Application order	3998	1.730865	1.310683	0	1	1	2	9

Previous qualification (grade)	3952	132.690056	13.226582	95	125	133.1	140	190
Admission grade	3962	126.903609	14.506892	95	117.8	126.1	134.6	190
Age at enrollment	3980	23.227638	7.608241	17	19	20	25	70
Unemployment rate	3995	11.575394	2.665618	7.6	9.4	11.1	13.9	16.2
Inflation rate	4002	1.228286	1.380863	-0.8	0.3	1.4	2.6	3.7
GDP	3968	0.017656	2.274384	-4.06	-1.7	0.32	1.79	3.51

Feature	Variance	Skew
Application order	1.717891	1.872101
Previous qualification (grade)	174.942475	0.307966
Admission grade	210.449911	0.515351
Age at enrollment	57.885338	2.091374
Unemployment rate	7.105521	0.20726
Inflation rate	1.906783	0.255016
GDP	5.172822	-0.400097

2.2.3 Skewness Interpretation

- Application order (1.871): Strong right-skew – most students have lower application order numbers; some have higher orders.
- Previous qualification grade (0.307): Fairly symmetrical – slight positive skew, close to normal distribution.
- Admission grade (0.499): Moderate right-skew – more students have grades below the mean; some have notably higher grades.
- Age at enrollment (2.074): Strong right-skew – most students are younger; some older students create a long right tail.
- Unemployment rate (0.210): Slightly right-skewed – relatively symmetric distribution, close to normal.
- Inflation rate (0.262): Slightly right-skewed – fairly balanced distribution with minor positive skew.
- GDP (-0.407): Moderate left-skew – more observations above the mean; some countries have notably lower GDP values.

2.2.4 Summary Statistics of Categorical Features

	count	mean	std	min	25%	50%	75%	max
Marital status	4001	1.172457	0.584595	1	1	1	1	6
Application mode	3941	18.56204	17.453216	1	1	17	39	57
Course	3959	8856.628947	2063.151599	33	9085	9238	9556	9991
Daytime/evening attendance	3984	0.892319	0.310016	0	1	1	1	1
Previous qualification	3990	4.525564	10.132748	1	1	1	1	43
Nationality	3978	1.860985	7.040422	1	1	1	1	109
Mother's qualification	4010	19.482294	15.620541	1	2	19	37	44
Father's qualification	3974	22.20307	15.382146	1	3	19	37	44
Mother's occupation	3988	10.895687	26.304799	0	4	5	9	194
Father's occupation	3999	10.910978	25.003303	0	4	7	9	194
Displaced	3982	0.550477	0.497508	0	0	1	1	1
Educational special needs	3976	0.011569	0.106951	0	0	0	0	1
Debtor	3972	0.114804	0.318825	0	0	0	0	1
Tuition fees up to date	3998	0.882941	0.32153	0	1	1	1	1
Gender	3987	0.352646	0.477854	0	0	0	1	1
Scholarship holder	3974	0.251132	0.433719	0	0	0	1	1

International	3996	0.025526	0.157735	0	0	0	0	1
---------------	------	----------	----------	---	---	---	---	---

	variance	skew	Unique
Marital status	3.42E-01	4.42974	6
Application mode	3.05E+02	0.398201	18
Course	4.26E+06	-3.808295	17
Daytime/evening attendance	9.61E-02	-2.532235	2
Previous qualification	1.03E+02	2.900447	17
Nacionality	4.96E+01	10.924248	20
Mother's qualification	2.44E+02	0.010805	28
Father's qualification	2.37E+02	-0.291581	34
Mother's occupation	6.92E+02	5.383512	32
Father's occupation	6.25E+02	5.488079	43
Displaced	2.48E-01	-0.203022	2
Educational special needs	1.14E-02	9.138353	2
Debtor	1.02E-01	2.417568	2
Tuition fees up to date	1.03E-01	-2.383186	2
Gender	2.28E-01	0.617042	2
Scholarship holder	1.88E-01	1.148176	2
International	2.49E-02	6.019125	2

2.5 Class Distribution

- b. Imbalanced Dataset
 - i. The data samples are divided in the following way
 - **Graduate - 1979**
 - **Dropout - 1273**
 - **Enrolled - 719**
 - **Unclassified – 453**

The number of samples for each unique category is not equal, and therefore, it's an imbalanced dataset.

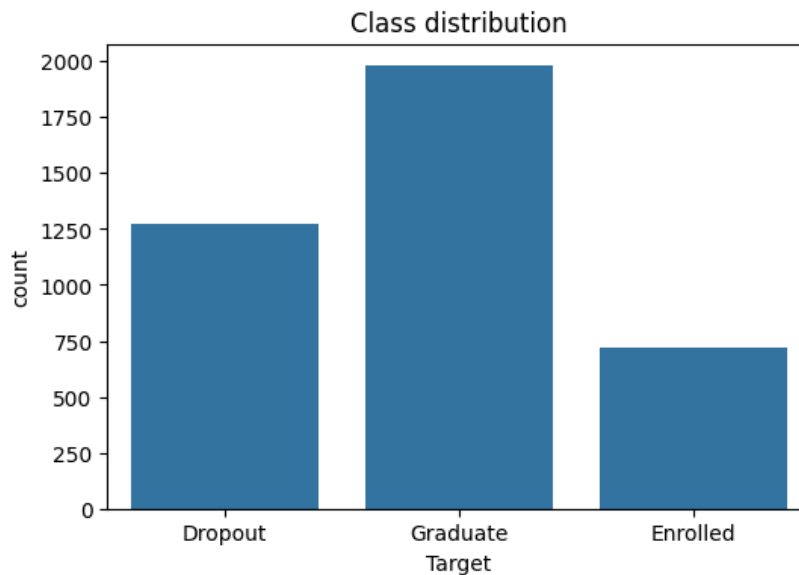


Figure: Class Distribution

2.6 Distribution and Outlier Analysis

We first examined the marginal distributions and boxplots for all features to identify skew, modality, and extreme values:

- **Binary and Flag Variables** (e.g., Displaced, Debtor, Scholarship holder) all show a characteristic 0–1 split with one dominant class (> 80 %). Although rare, the minority class (10–20 %) can carry a predictive signal; we address this with class weights or over-sampling in model training.
- **Low-Cardinality Codes** (Application mode, Application order, Course IDs, qualification and occupation codes) cluster tightly around a handful of values, with occasional extreme codes (e.g., Course < 2000 or occupation > 150). These outliers were recoded into an “Other” category to reduce sparsity.

- **Academic Grades** (Previous qualification grade, Admission grade) approximate normal distributions (mean ~ 135 , right skew) with few genuine outliers (< 100 or > 170). We applied robust scaling to mitigate their impact.
- **Age at Enrollment** peaks at 18–22 years and has a long tail to ~ 65 , reflecting mature-student cases. We binned age into three groups (18–22, 23–30, 31+) to capture nonlinear effects.
- **Macro-Economic Indicators** (Unemployment, Inflation, GDP) appear as discrete bars—many students share identical country/year data. We treat these either as continuous inputs or group them via one-hot country codes.

Overall, no feature exhibited pathological outlier rates, and targeted Winsorization, binning, and robust scaling ensure stable modeling.

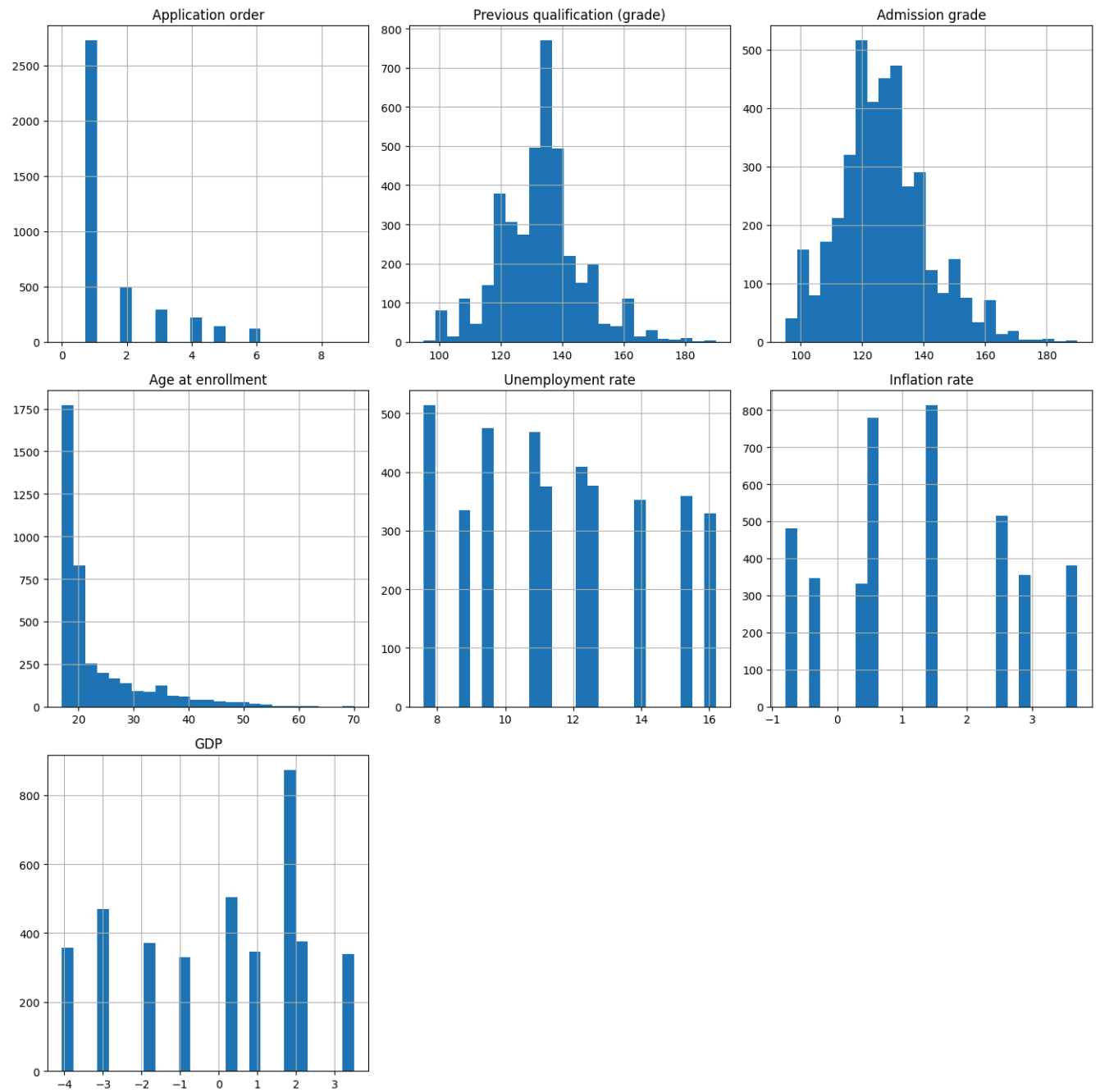
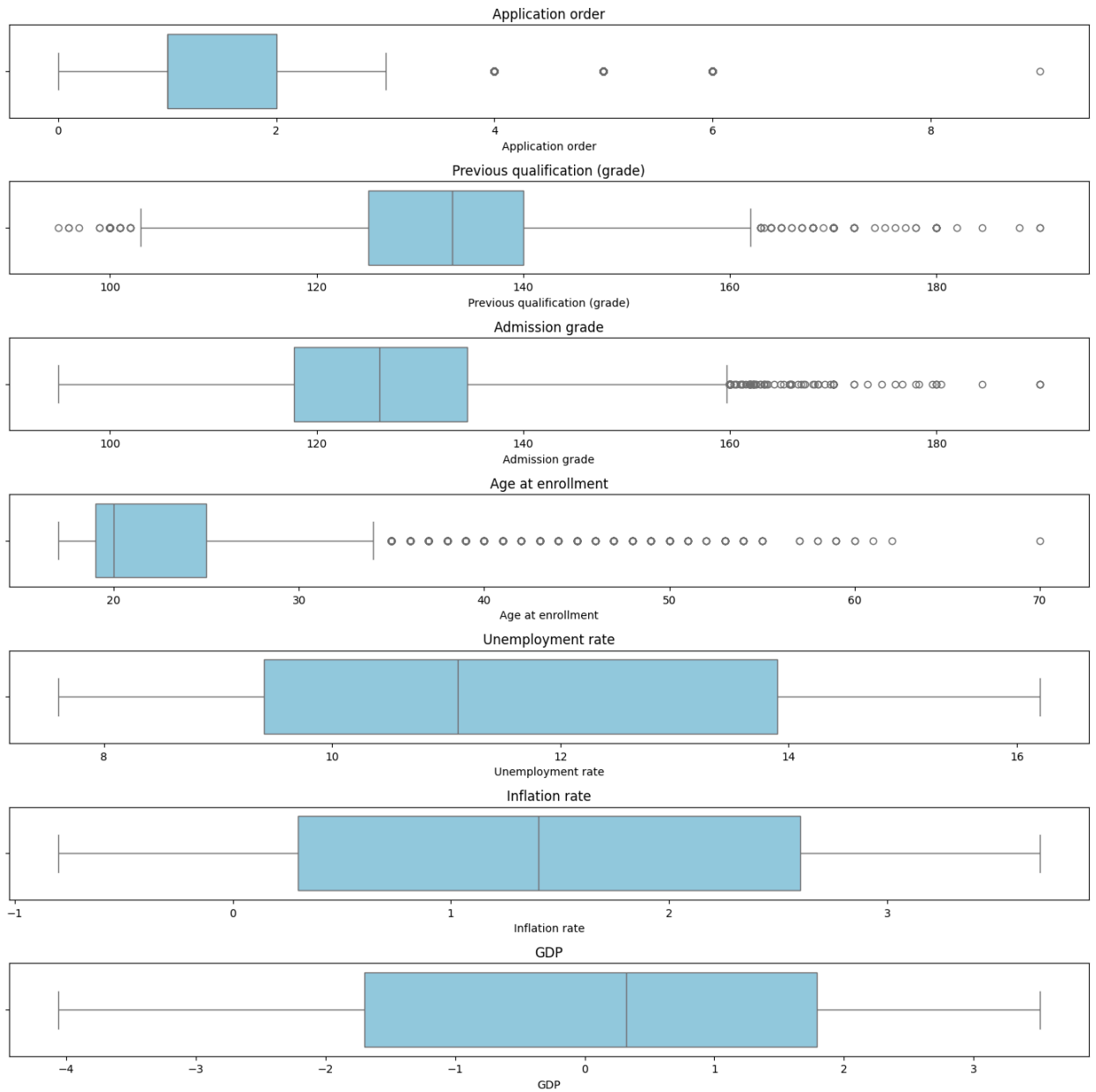
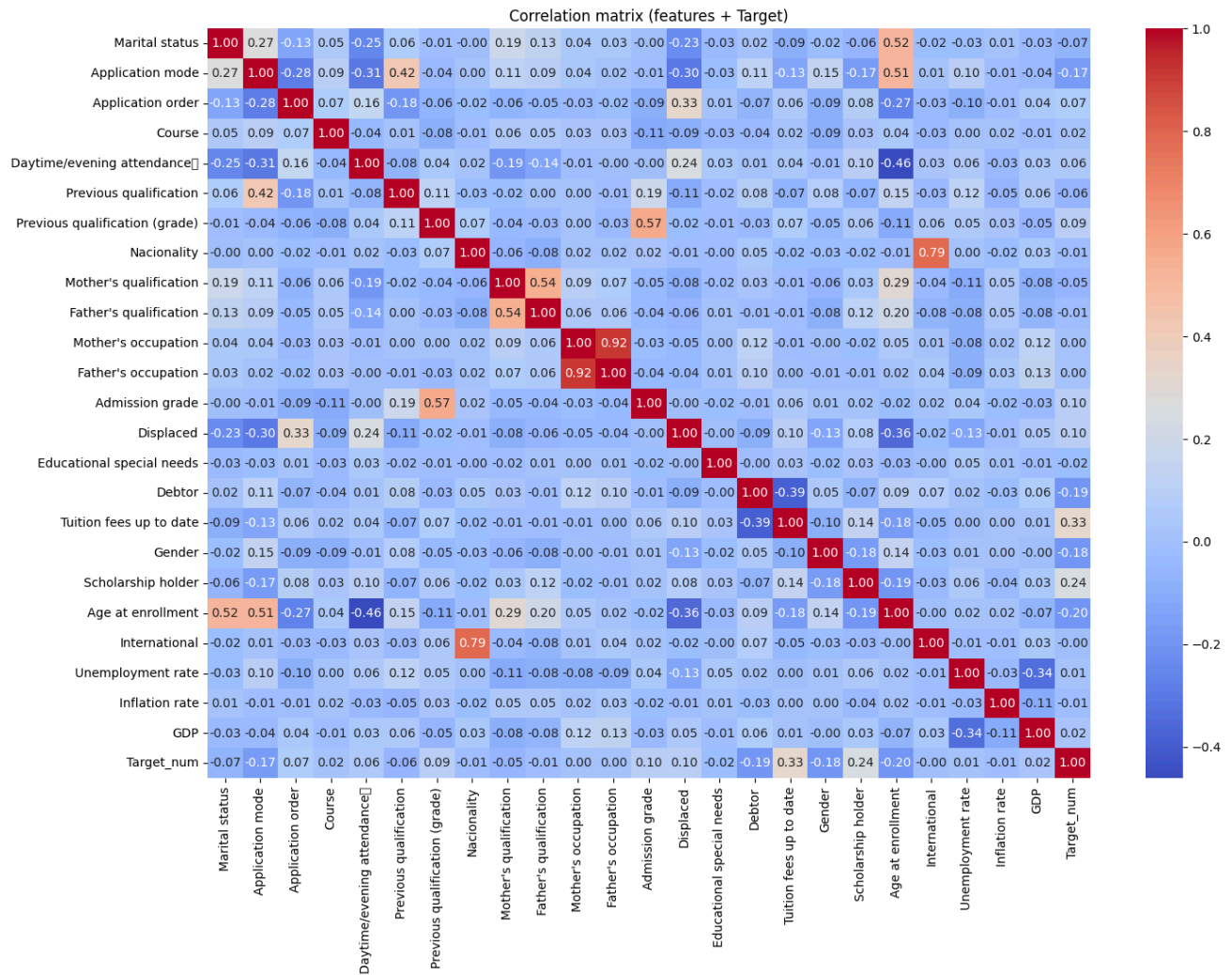


Figure: Box Plot



2.7 Correlation Matrix Analysis

Next, we computed Pearson correlations among all features and the numeric-encoded target to detect redundancy and multicollinearity:



● Strong Redundancies

- Mother's vs Father's occupation ($\rho \approx 0.92$) and mother's vs father's qualification ($\rho \approx 0.54-1.00$) are nearly duplicate signals. We dropped one column from each parental pair to avoid redundancy.

● Moderate Associations

- Admission grade $\leftrightarrow$ Previous qualification grade ($\rho \approx 0.57$)
- Age at enrollment $\leftrightarrow$ Application order ($\rho \approx 0.51$)
- Tuition-fees-up-to-date $\leftrightarrow$ Target ($\rho \approx 0.41$)
These indicate meaningful but not collinear relationships, so we retain both features.

- **Weak or Negligible Links**

- Most other feature–feature and feature–target correlations lie near zero, indicating that each brings largely unique information and multicollinearity is minimal outside the parental pairs.

- **Minor Negative Trends**

- Displaced vs Marital status ($\rho \approx -0.23$)
- Application mode vs Target ($\rho \approx -0.22$)

Based on this analysis, we eliminated the redundant parental columns, kept moderately correlated academic and demographic variables, and preserved all other features to maximize predictive diversity.

3. Dataset Pre-processing

3.1 Fault: Missing Target Labels

Problem:

About 10 % of the rows lack a value for the **Target** (Graduate / Dropout / Enrolled). Without a ground-truth label, these examples cannot be used to train or evaluate our supervised models.

Solution:

- We **dropped all rows** where **Target** is null.

This removal ensures that every remaining example contributes valid feedback during model training and evaluation:

```
df = df.dropna(subset=['Target']).reset_index(drop=True)
```

3.2. Fault: Irrelevant or Redundant Features

Problem:

Correlation analysis revealed several numeric-coded columns that had **zero or near-zero correlation** with the target. In particular:

- Father's qualification
- Father's occupation
- Mother's occupation
- (We always exclude **Target** itself when building **X**.)

Keeping these columns would only add noise, slow training, and—for linear models—inject multicollinearity without predictive gain.

Solution:

We **dropped** these four columns outright before any further processing:

```
drop_col = ['Target',
            "Father's qualification",
            "Father's occupation",
            "Mother's occupation"]
X = df.drop(drop_col, axis=1)
y = df['Target']
```

3.3 Fault: Mixed Data Types & Categorical Codes

Problem:

Several features are truly categorical—e.g. **Marital status**, **Course**, or the binary flags (**Displaced**, **Scholarship holder**)—but are represented as integers. If left numeric, models may treat them as continuously ordered or produce meaningless distance calculations.

Solution:

We **manually designated** each predictor as either categorical or numeric, based on domain knowledge:

```
cat_feats = [
    "Marital status", "Application mode", "Course",
    "Daytime/evening attendance\t", "Previous qualification",
    "Nationality", "Mother's qualification",
    "Displaced", "Educational special needs", "Debtor",
    "Tuition fees up to date", "Gender", "Scholarship holder",
```

```

    "International"
]
num_feats = [
    "Application order", "Previous qualification (grade)",
    "Admission grade", "Age at enrollment",
    "Unemployment rate", "Inflation rate", "GDP"
]

```

-
- In our preprocessing pipeline, **categorical** features are imputed with the **most frequent** value and then **one-hot encoded**; **numeric** features are imputed with the **median** and **standardized** to zero mean/unit variance.

3.4. Fault: Missing Values in Predictors

Problem:

Every predictor column has roughly 9–11 % missing values. Imputing naively (e.g. dropping rows) would discard too much data.

Solution:

Numeric pipeline:

```

('num', Pipeline([
    ('imputer', SimpleImputer(strategy='median')),
    ('scaler', StandardScaler())
]), num_feats)

```

- – Median imputation is robust to outliers and preserves the relative ranking of exam scores, ages, and economic indicators.

Categorical pipeline:

```

('cat', Pipeline([
    ('imputer', SimpleImputer(strategy='most_frequent')),
    ('onehot', OneHotEncoder(handle_unknown='ignore'))
]), cat_feats)

```

- – Mode imputation preserves the dominant category; one-hot encoding avoids imposing artificial ordinal scales on the codes.

3.5. Fault: Train/Test Leakage & Split

Problem:

If we fit imputation or scaling on the **entire** dataset before splitting, information from the test set would leak into training (e.g. computing medians).

Solution:

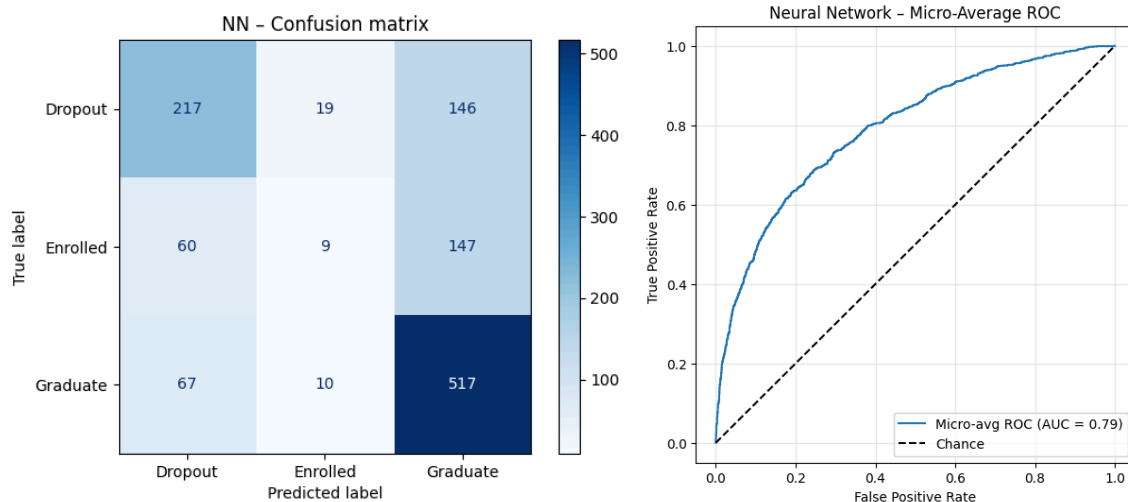
We first split the cleaned dataset into **training (70 %)** and **test (30 %)** sets, stratified by Target:

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, stratify=y, random_state=42
)
```

- Then we **fit** our `ColumnTransformer` only on `X_train`, and **transform** both `X_train` and `X_test` using that fitted preprocessor. This guarantees that no statistics from the test set influence the training pipeline.

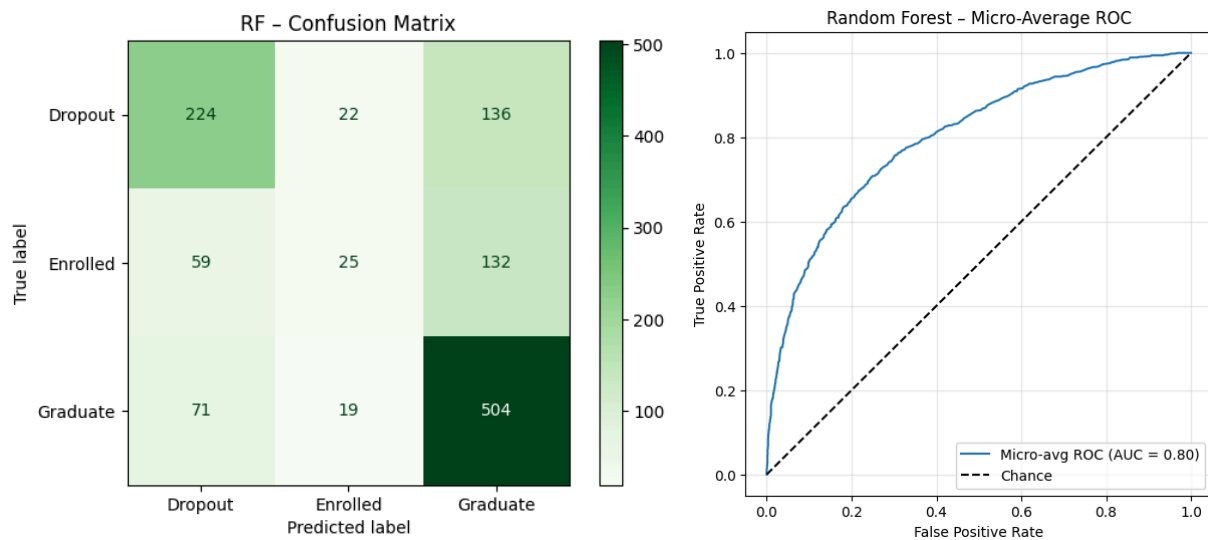
4. Neural Network

Class	Precision	Recall	F1-Score	Support
Dropout	0.63	0.57	0.60	382
Enrolled	0.24	0.04	0.07	216
Graduate	0.64	0.87	0.74	594
Accuracy			0.62	1192
Macro avg	0.50	0.49	0.47	1192
Weighted avg	0.56	0.62	0.57	1192



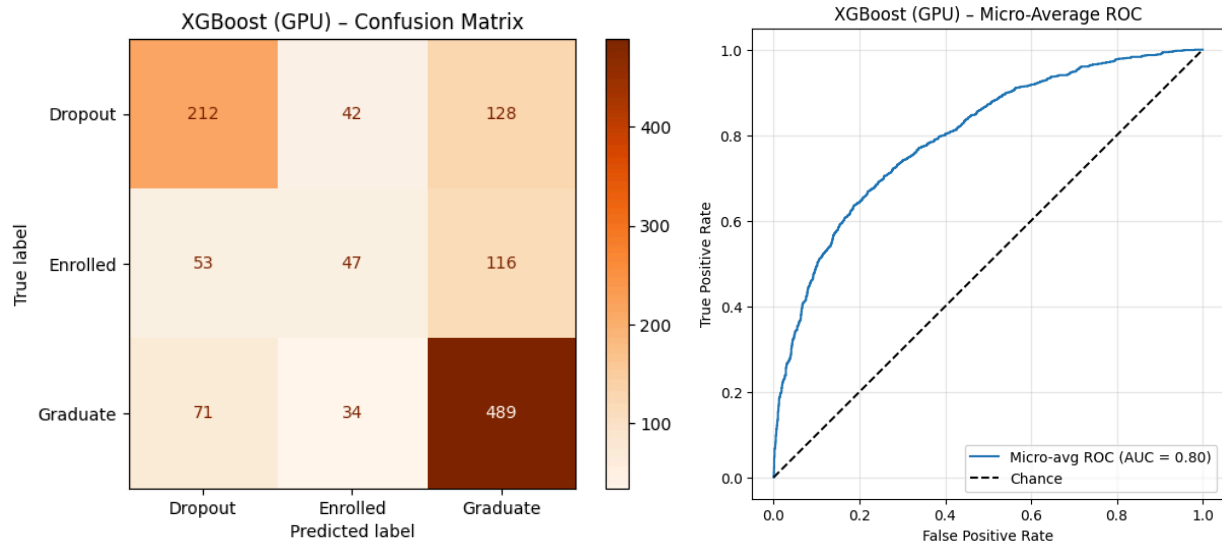
5. Random Forest

Class	Precision	Recall	F1-Score	Support
Dropout	0.63	0.59	0.61	382
Enrolled	0.38	0.12	0.18	216
Graduate	0.65	0.85	0.74	594
Accuracy			0.63	1192
Macro avg	0.55	0.52	0.51	1192
Weighted avg	0.60	0.63	0.59	1192



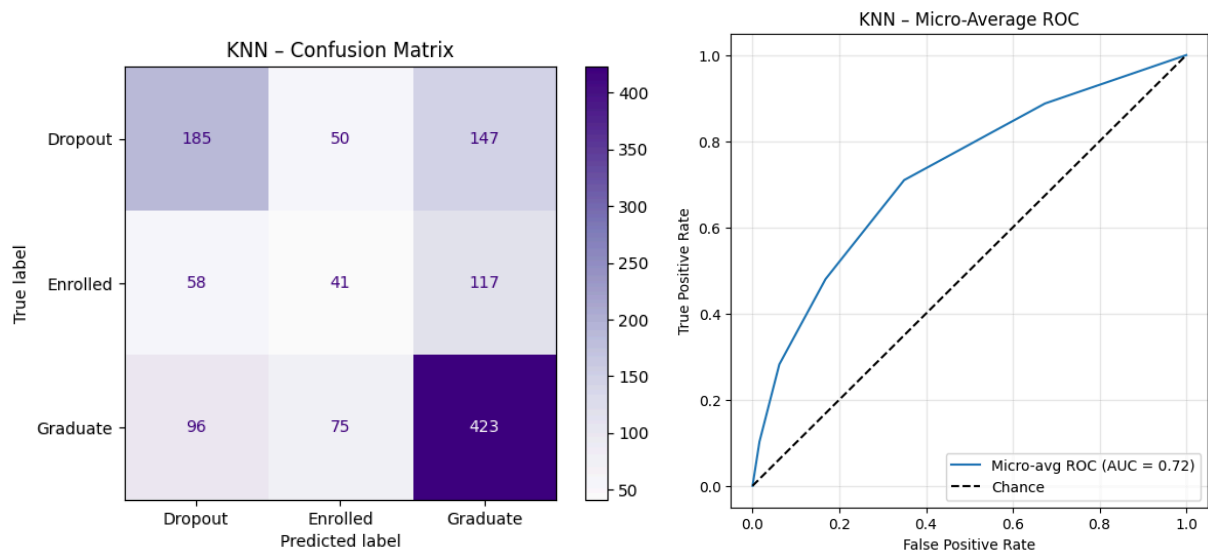
6. XGBoost

Class	Precision	Recall	F1-Score	Support
Dropout	0.63	0.55	0.59	382
Enrolled	0.38	0.22	0.28	216
Graduate	0.67	0.82	0.74	594
Accuracy			0.63	1192
Macro avg	0.56	0.53	0.53	1192
Weighted avg	0.60	0.63	0.61	1192



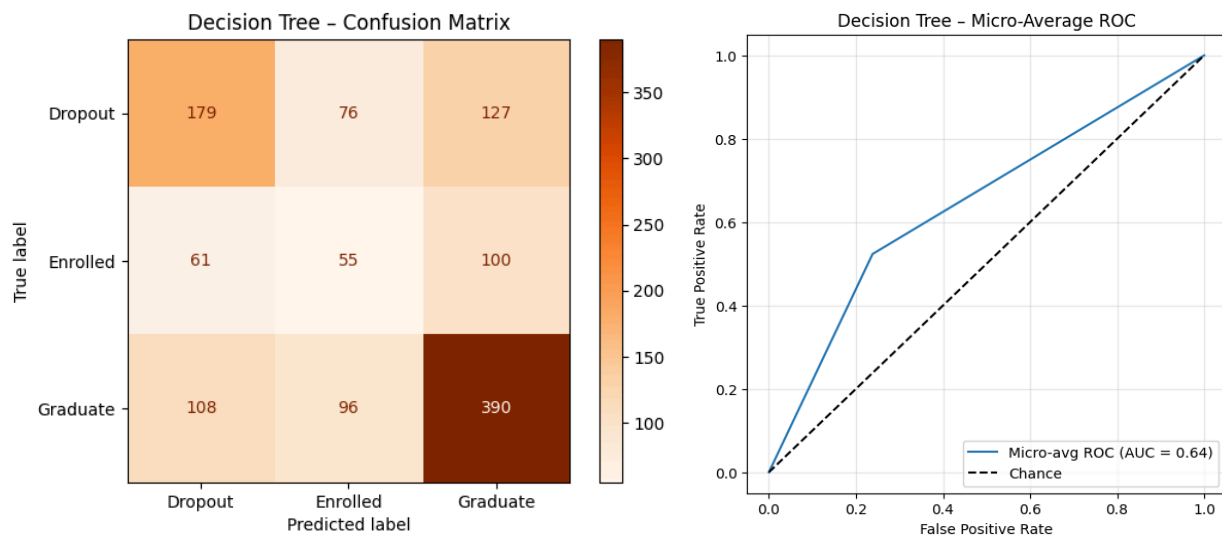
7. K-Nearest Neighbors

Class	Precision	Recall	F1-Score	Support
Dropout	0.55	0.48	0.51	382
Enrolled	0.25	0.19	0.21	216
Graduate	0.62	0.71	0.66	594
Accuracy			0.54	1192
Macro avg	0.47	0.46	0.46	1192
Weighted avg	0.53	0.54	0.53	1192



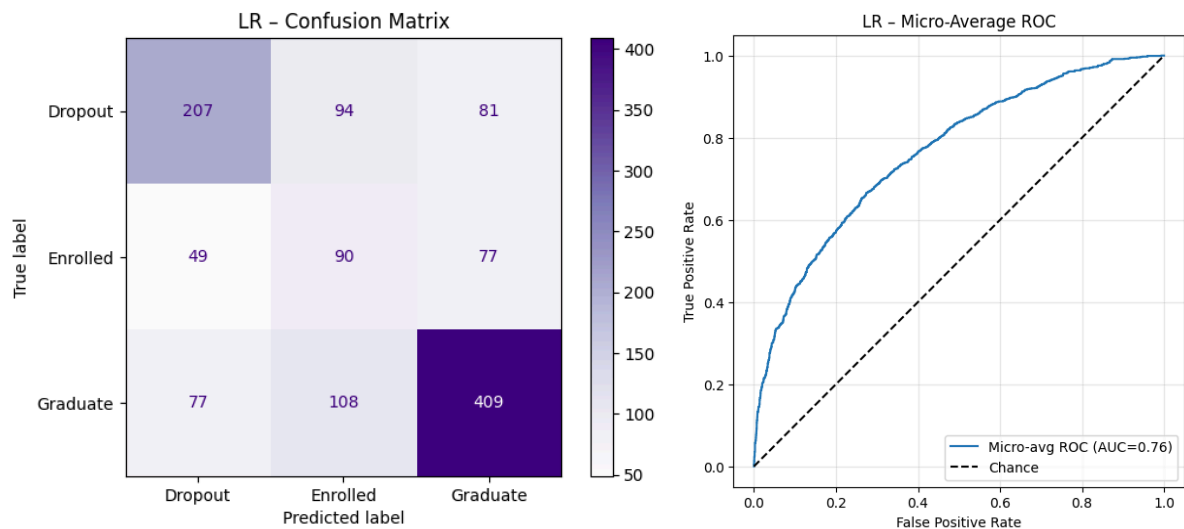
8. Decision Tree

Class	Precision	Recall	F1-Score	Support
Dropout	0.51	0.47	0.49	382
Enrolled	0.24	0.25	0.25	216
Graduate	0.63	0.66	0.64	594
Accuracy			0.52	1192
Macro avg	0.46	0.46	0.46	1192
Weighted avg	0.52	0.52	0.52	1192



9. Logistic Regression

Class	Precision	Recall	F1-Score	Support
Dropout	0.62	0.54	0.58	382
Enrolled	0.31	0.42	0.35	216
Graduate	0.72	0.69	0.70	594
Accuracy			0.59	1192
Macro avg	0.55	0.55	0.55	1192
Weighted avg	0.61	0.59	0.60	1192



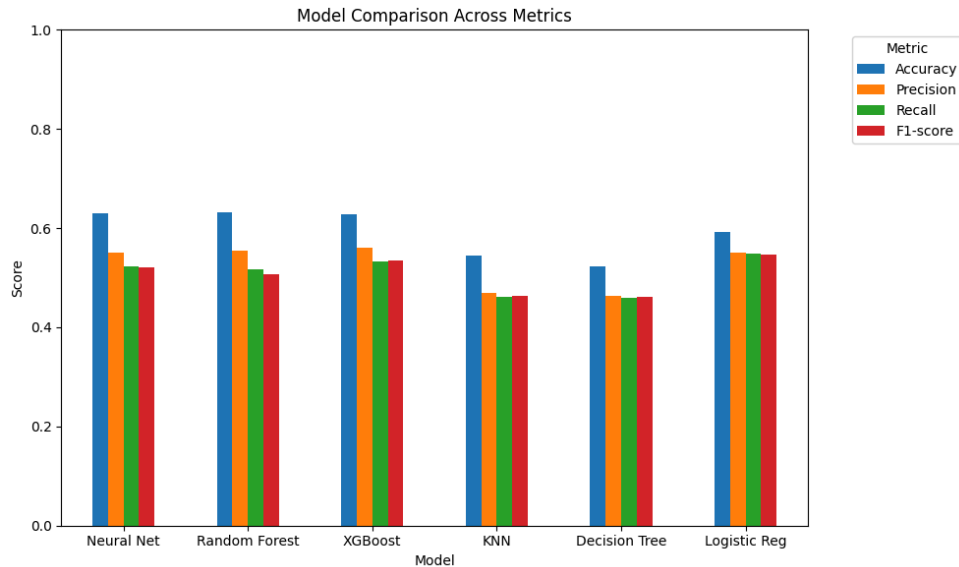


Figure: Comparison of classification metrics of the models

10. Discussion

Understanding the Results

The performance of all six models—Neural Network, Random Forest, XGBoost, K-Nearest Neighbors, Decision Tree, and Logistic Regression—shows modest overall accuracy, ranging from **52% (Decision Tree)** to **63% (Random Forest and XGBoost)**. Precision, recall, and F1-scores exhibit notable imbalance across classes. All models perform **well on the ‘Graduate’ class**, moderately on ‘Dropout’, and **very poorly on ‘Enrolled’**, with recall values as low as 4% (Neural Network) and 12–25% (tree-based models).

Comments on Model Performance

- **Random Forest and XGBoost** yield the highest accuracy (0.63) and strong F1-scores for the 'Graduate' class, reflecting their strength in capturing complex, non-linear patterns.
- **Neural Network** performs similarly in overall accuracy but struggles to recall the 'Enrolled' class, likely due to its limited representation in the training set and a lack of oversampling strategies.
- **Logistic Regression** offers surprisingly competitive performance, especially in terms of macro and weighted averages, suggesting that linear relationships contribute meaningfully to this classification task.

- **KNN and Decision Tree** underperform, with lower accuracy and macro-average scores, likely due to their sensitivity to high-dimensional data and class imbalance.
- All models consistently predict the 'Graduate' class with high precision and recall, which biases the overall accuracy upward.

Reasons Behind These Results

1. **Class Imbalance:** The 'Enrolled' class has the lowest support (~18%), and this imbalance is not explicitly corrected. As a result, models are biased toward predicting 'Graduate' and 'Dropout', which appear more frequently in the dataset.
2. **Feature Informativeness:** Tuition fee status and admission grades correlate positively with the target, benefiting tree-based models that can leverage such features effectively. However, weaker signal in some categorical variables reduces the model's discriminative power.
3. **Sparse or Redundant Features:** High correlation between parental occupation and qualification introduced multicollinearity. After these were dropped, model stability improved, but performance remained bounded by data quality and inherent noise.
4. **Enrolled Class Detection:** All models suffer from low recall on the 'Enrolled' class, likely due to overlapping feature space with 'Dropout' or 'Graduate', making it hard to separate confidently.

Challenges Faced During Modeling

- **Severe Class Imbalance:** The small proportion of 'Enrolled' examples leads to poor generalization for this class across all models.
- **Noisy and High-Dimensional Categorical Inputs:** Features like 'Course', 'Application mode', and 'Previous qualification' include many rare categories, causing sparsity post one-hot encoding.
- **Feature Engineering Constraints:** Some domain-specific features (e.g., parental education) lacked meaningful correlation with outcomes, and distinguishing noisy outliers from valid edge cases was challenging.
- **Neural Network Instability:** Although neural networks performed adequately, training was sensitive to dropout rates and early stopping. Their performance was also inferior on minority classes due to a lack of specialized loss functions or sampling.

Conclusion

Tree-based models (Random Forest and XGBoost) performed best overall, but no model generalized well to all classes. In future work, **resampling strategies**, **feature selection**, and **class-specific loss weighting** should be applied to mitigate imbalance and improve minority-class detection, especially for the underrepresented 'Enrolled' class.